

COMPLETE MIXED-TRACE FIBRES AND SHARP HARD-PRIME MIDDLE SCALES

THE CLANKERS

ABSTRACT. We complete a source fibre left unclassified by the corrected modulo-24 Erdős–Straus trace-return map, and prove that three established inverse bounds for its literal receiver are exponent-sharp already on actual middle witnesses with prime $p \equiv 1 \pmod{840}$. Neither result supplies an initial witness at every prime. Complete higher-order coefficients, a scalar observation hierarchy, and independent finite checks are in the accompanying workbench proof.

1. THE BARE SORTED FIBRE

Fix a prime $p \equiv 1 \pmod{24}$. The original mixed rational source has $a_0 = cq$, $c \mid 6$, $q > 3$ prime, $p/4 < a_0 < p/2$, and positive rational tails whose reciprocal sum with $1/a_0$ is $4/p$ and whose sum is an integer. With $R_0 = 4a_0 - p$, every such pair has an original word $u_0 \mid a_0^2$, normalized as $a_0 = h_0 r_0 s_0$, $u_0 = h_0 r_0^2$, $(r_0, s_0) = 1$. Its E tails are $(h_0 s_0 \kappa, ph_0 r_0 \kappa)$, $\kappa = (pr_0 + s_0)/R_0$, with a separate swap bit; its M tails are $(ph_0 s_0 \lambda, ph_0 r_0 \lambda)$, $\lambda = (r_0 + s_0)/R_0$. The squared gates are respectively $R_0 \mid (4u_0 + 1)^2$ and $R_0 \mid (p + 4u_0)^2$. The factorization $(R_0 y - pa_0)(R_0 z - pa_0) = (pa_0)^2$ proves completeness, including both orders. The coordinate antecedents are Type I/II (Elsholtz and Tao, 2013).

The corrected source-changing map (The Clankers, 2026a,b) writes $R_0 = \delta t^2$ with δ its squarefree part. E inputs have $r_0 s_0 \mid 6$ and return the middle state

$$H = (\delta + 1)/(4r_0 s_0), \quad J = (pr_0 + s_0)/\delta, \quad (H, s_0, J, r_0).$$

M inputs are complemented if necessary to a word $\nu \mid c^2 \mid 36$; they return that word at $a = (p + \delta)/4$. Sorting forgets the orientations. This map is denoted by ρ .

Let an existing sorted M target be

$$W = (p; a, Y, Z) = (p; hrs, phr\lambda, phs\lambda), \quad r < s, \quad R = 4a - p = (r + s)/\lambda.$$

These primitive coordinates are unique: reduce $(Y/p, Z/p)$ by its gcd to get (r, s) , then $h = a/(rs)$ and $\lambda = \gcd(Y/p, Z/p)/h$. Write $\text{Car}(A)$ for the literal, uniquely decoded condition $A = cq$, $c \mid 6$, $q > 3$ prime. For $n = \prod_{\ell} \ell^{e_{\ell}}$ put $K(n) = \prod_{\ell} \ell^{\lfloor e_{\ell}/2 \rfloor}$.

Theorem 1. *The complete ordered fibre $\rho^{-1}(W)$ is as follows. Put $\delta_E = 4hr\lambda - 1$. If $r\lambda \mid 6$ and δ_E is squarefree, then for every*

$$t \mid \text{oddpart}(s), \quad \delta_E t^2 < p, \quad a_t = (p + \delta_E t^2)/4 \text{ satisfying } \text{Car},$$

there are two E inputs, with both orders, and

$$(h_t, r_t, s_t) = (a_t/(r\lambda), \lambda, r), \quad u_t = h_t \lambda^2.$$

There are no other E inputs. For every $\nu \in \{hr^2, hs^2\} \cap \text{Div}(36)$, if R is squarefree, put $A_{\nu} = (p + 4\nu)/R$. Every

$$t \mid A_{\nu}, \quad Rt^2 < p, \quad a_t = (p + Rt^2)/4 = c_t q_t, \quad c_t \mid 6, \quad q_t > 3 \text{ prime}, \quad \nu \mid c_t^2$$

gives exactly the two M words $\nu, a_t^2/\nu$. There are no other M inputs. All lists are disjoint; hence their cardinalities add with the displayed factor two.

Proof. For E, the normalized target forces $(r_0, s_0, H, J, \delta) = (\lambda, r, h, s, \delta_E)$. The reconstructed coordinates satisfy

$$r(4u_t + 1) = \delta_E(s + \lambda t^2), \quad \gcd(r, \delta_E t^2) = 1.$$

Thus the squared gate is equivalent to $\delta_E t^2 \mid (\delta_E s)^2$, and this is equivalent primewise to $t \mid \text{oddpart}(s)$ because δ_E is squarefree and the residual is odd. Since $\lambda R = r + s$ and $(r, s) = 1$, one has $(s, r\lambda) = 1$. Hence $t \mid \text{oddpart}(s)$ gives $(t, 4r\lambda) = 1$. Every such unit has square one modulo the divisor $4r\lambda$ of 24, so multiplication by t^2 preserves the source shell condition. The displayed original word is therefore available and its forward return is W . The carrier test is exactly the mixed-source restriction.

For M, the sorted target has only the two word orientations hr^2, hs^2 . For these two choices respectively,

$$A_{hr^2} = 4hr\lambda - 1, \quad A_{hs^2} = 4hs\lambda - 1,$$

so $A_\nu = (p + 4\nu)/R$ is a positive odd integer. The selected small word is one of them and divides c_t^2 . Its old squared gate is $Rt^2 \mid (RA_\nu)^2$, exactly $t \mid A_\nu$ when R is squarefree. Moreover $A_\nu \equiv -1 \pmod{4K(\nu)}$, and complementation from u_t to $u'_t = a_t^2/u_t$ preserves the square gate because

$$p + 4u'_t \equiv a_t u_t^{-1} (p + 4u_t) \pmod{Rt^2}$$

with a unit multiplier. Indeed, $\gcd(a_t, Rt^2) = 1$ because a common divisor would divide $4a_t - Rt^2 = p$, and $\gcd(u_t, Rt^2) = 1$ follows from $u_t \mid a_t^2$. The original square budget and the gate are therefore retained. The two source words have carrier exponent zero and two, hence cannot coincide. The carrier is unique, so different small words cannot duplicate a source. E and M inputs have different p -valuation pairs in the original factor coordinates. These same equations give the complete inverse in every case. $\square$

The E general-ray section $t = 1$ does not always survive the carrier restriction. The entire fibre over $(1009; 255, 24216, 686120)$ consists of the two M words 9, 11449 at $a_t = 321 = 3 \cdot 107, t = 5$. Its squarefree $a = 255$ is not a mixed carrier. At $(6841; 1866, 20523, 12765306)$, two E orders from $a_t = u_t = 1713$ and two M words 3, 1160652 from $a_t = 1866$ collide. The free-module kernel over a nonempty undecorated fibre has its literal difference basis $e_T - e_{T_0}$. After adding the sixteen receiver frames, its rank is multiplied by sixteen. Neither statement proves a fibre nonempty.

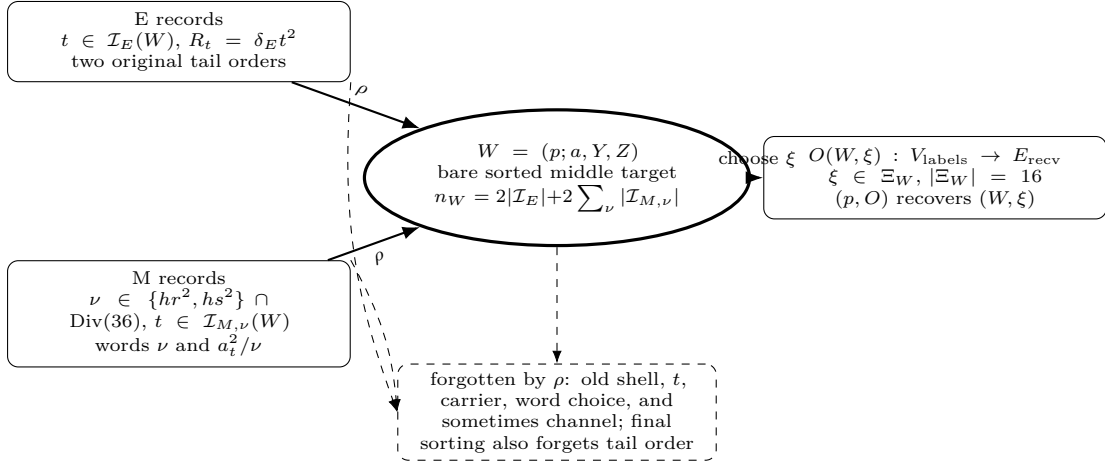


FIGURE 1. The exact arithmetic fibre and receiver lift. The E and M records are disjoint as source records but may meet at one target. Adding the full matrix restores W and the selected square-root frame, not the forgotten arithmetic record. The fibre theorem and receiver reconstruction formulas give the exact maps.

For the receiver below, its full matrix and p reconstruct the target:

$$\xi_j = O_{0j}, \quad d_j = iO_{1j}/O_{0j}, \quad A = (O_{20}/O_{00} - 2pd_0)/d_0^2, \quad \ell_j = (O_{2j}/O_{0j} - Ad_j^2)/(2d_j).$$

On the marked middle domain $a < p < Y < Z$, all four roots are distinct, $d_j = -S^{-1} \prod_{k \neq j} (\ell_j - \ell_k) \neq 0$, and $\xi_j \neq 0$. Thus every displayed denominator is nonzero. Direct substitution proves both compositions on the receiver image. The same arithmetic fibre is therefore the complete fibre of the fixed-prime receiver-valued map for each of the sixteen independently selected square-root frames.

2. MIDDLE FAMILIES IN THE HARD CLASS

The original literal receiver is the typed map $O : V_{\text{labels}} \rightarrow E_{\text{recv}}$ with

$$H(X) = -S^{-1} \prod_{j=0}^3 (X - \ell_j), \quad \ell = (p, a, Y, Z), \quad S = \sum_j \ell_j, \quad A = -1/S,$$

$$d_j = H'(\ell_j), \quad \xi_j^2 d_j = 1, \quad O_{\cdot j} = \xi_j (1, -id_j, Ad_j^2 + 2\ell_j d_j, i(7\ell_j^2 d_j - 13d_j^2))^T.$$

The established uniform bounds on existing integral witnesses are $O(p^{3/2}), O(1), O(p^{-2})$ for the first three inverse compounds (The Clankers, 2026c, Theorem 1.1). Norms are the original standard Hermitian norms.

Theorem 2. *All three uniform powers are optimal already on middle witnesses at primes $p \equiv 1 \pmod{840}$.*

Proof. First fix $R = 11, u = 9$, and take primes $p \equiv 8401 \pmod{9240}$. Put

$$a = (p + R)/4, \quad Y = p(a + u)/R, \quad Z = pa(a + u)/(Ru).$$

Here $9 \mid a^2$ and $11 \mid p + 36$, so these are original integer M states. More generally fix positive R, u , and let $\alpha = 1/(4R)$, $\beta = 1/(16Ru)$ and $C = \sqrt{3/(16R)}$. Choose positive real ξ_a, ξ_Y and positive imaginary ξ_p, ξ_Z . The exact rational roots give

$$d_p \sim -C^2 p^3, \quad d_a \sim C^2 p^3, \quad d_Y \sim \alpha^2 p^4, \quad d_Z \sim -\beta^2 p^6.$$

The compensated analytic matrix

$$B = \text{diag}(1, p^{-3}, p^{-4}, p^{-6}) O \text{diag}(p^{3/2}, p^{3/2}, 1, p^{-3})$$

has limit

$$B_0 = \begin{pmatrix} i/C & 1/C & 0 & 0 \\ -C & -iC & 0 & 0 \\ -2iC & C/2 & 2\alpha^2 & 0 \\ 13C^3 & -13iC^3 & -6i\alpha^3 & 20\beta^3 \end{pmatrix}.$$

The four leading compound norms are

$$20\beta^3 p^9, \quad 40\alpha^2 \beta^3 p^{13}, \quad 40\sqrt{2}C\alpha^2 \beta^3 p^{29/2}, \quad 80\alpha^2 \beta^3 p^{13}.$$

They use respectively the entry $(3, Z)$, the minor $(2, 3; Y, Z)$, both triples $(1, 2, 3; p, Y, Z)$ and $(1, 2, 3; a, Y, Z)$, and the full determinant. Every other compound entry is a lower power, as follows by adding row powers $(0, 3, 4, 6)$ and column powers $(-3/2, -3/2, 0, 3)$ and retaining zero leading minors. The complete leading row and the Frobenius bound therefore sandwich the true compound norms. Consequently

$$p^{-3/2} \|O^{-1}\| \rightarrow \sqrt{3/(32R)}, \quad \|\wedge^2 O^{-1}\| \rightarrow \frac{1}{2}.$$

These nonzero limits settle the first two powers.

For the third, take primes $p \equiv 4201 \pmod{9240}$ and set $s = (p + 1)/11$, $(a, Y, Z) = (3s, 3p, 3ps)$. The original M coordinates are $(h, r, s, \lambda) = (3, 1, s, 1)$. The small root ratios are $x = (1, 3/11, 3)$, their derivative constants are $D = (-16/11, 240/121, 60/11)$, and $Z \sim (3/11)p^2$. The leading core matrix in rows $0, 1, 2$ is obtained by multiplying each column $(1, -iD_j, 2x_j D_j)^T$ by the selected inverse square root of D_j . Its determinant is $2896/121$. The last row of O on the Z column is $20(3/11)^3 p^6 [1 + O(p^{-1})]$; eliminating that column affects only lower core powers. The full determinant is therefore that entry times $(2896/121)p^2 [1 + O(p^{-1})]$. Since $\|\wedge^3 O^{-1}\| = \|O\|/|\det O|$,

$$p^2 \|\wedge^3 O^{-1}\| \rightarrow 121/2896 > 0.$$

Both residues are coprime to 9240 and are $1 \pmod{840}$. Dirichlet's theorem supplies infinitely many prime members (Dirichlet, 1837); see also Sutherland (2017, Theorem 18.1). No simultaneous primality of an additional carrier, nor universal occupancy of the mixed trace source, is asserted. $\square$

The complete proof gives all first singular corrections, not just rates:

$$6R + 14u/5, \quad -R/2 + 19u/5, \quad -41R/12, \quad 11R/12.$$

It also proves the middle scalar heat-observation hierarchy $0, 2, 3, 9$, retaining the small last coordinate, original signs, and metric factors. Those refinements are not needed for the theorem just proved. The new

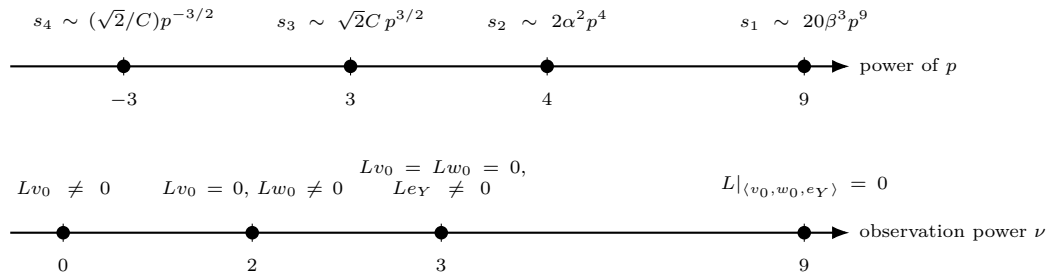


FIGURE 2. Two different exact exponent systems for the fixed- (R, u) middle family. The upper axis records the four singular-value scales. The lower axis records the exhaustive rescalings of $L \exp(-\tau p^3 O^* O) L^*$ for fixed $\tau > 0$; its last direction is e_Z . The first corrections and coefficients are stated in the middle-spectrum theorem and the subsequent observation table.

fixed-residual family is not the already published hard-prime exterior $c = 29$ family in ES16–ES17 (The Clankers, 2026d).

The executable standard-library checks include all 143 primes $p \equiv 1 \pmod{24}$ through 10000, all 1955 sorted M targets, all 1134 ordered mixed sources and every empty or occupied formula fibre. There are 490 image targets, 28 cross-channel fibres, and 156 image targets with no $t = 1$ input. These finite counts corroborate the implementations; they are not a universal existence theorem or independent human review.

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